NAME:K.V.SAI KRISHNA

REG.NO:192325037

Ex 6: Vacuum Cleaner Problem

CODE:

```
def reflex_vacuum_agent(location, status):
  if status == 'DIRTY':
    return 'SUCK'
  elif location == 'A':
    return 'RIGHT'
  else:
    return 'LEFT'

env = {'A': 'DIRTY', 'B': 'DIRTY'}
agent_loc = 'A'

print(f"Initial State: {env}")
for _ in range(4):
  action = reflex_vacuum_agent(agent_loc, env[agent_loc])
  print(f"Location: {agent_loc} | Status: {env[agent_loc]} -> Action: {action}")
  if action == 'SUCK':
    env[agent_loc] = 'CLEAN'
  elif action == 'RIGHT':
    agent_loc = 'B'
  elif action == 'LEFT':
```

```
    agent_loc = 'A'

print(f"Final State: {env}")
```

OUTPUT:

```
============== RESTART: C:/Users/Saiko/Documents/CSA1738/EXP-6.py =========
Initial State: {'A': 'DIRTY', 'B': 'DIRTY'}
Location: A | Status: DIRTY -> Action: SUCK
Location: A | Status: CLEAN -> Action: RIGHT
Location: B | Status: DIRTY -> Action: SUCK
Location: B | Status: CLEAN -> Action: LEFT
Final State: {'A': 'CLEAN', 'B': 'CLEAN'}
```